

Practice Material for Final Exam, Fall 2020, 46-921

1. If $X_1, X_2, \dots, X_n$ are iid $\text{Exponential}(\theta)$, what is the method of moments estimator for θ ?
2. Which of the following statements is true regarding the maximum likelihood estimator (MLE) for an unknown parameter θ ?

Choose One:

- (a) The MLE is always unbiased.
- (b) The MLE can only be found in cases where the model assumes that the random variables $X_1, X_2, \dots, X_n$ are iid from some known distribution.
- (c) There are conditions under which the mean squared error (MSE) of the MLE is minimal in an asymptotic sense, but there are situations in which, for fixed sample size n , there exists another estimator which has smaller MSE than the MLE.
- (d) The MLE is always approximately normal when the sample size n is large.

Questions 3 and 4 refer to the output below.

First, simulate 100 draws from a standard normal distribution.

```
In [71]: from scipy.stats import norm
import numpy as np

x = norm.rvs(loc=0, scale=1, size=100)
```

Next, imagine we fit a model which assumes the data are drawn from a Normal distribution with mean μ and variance σ^2 , and both μ and σ^2 are unknown.

The code below shows the MLE for each parameter being calculated, along with the loglikelihood for that fit.

```
In [72]: muhat1 = np.mean(x)
sigmahat1 = np.std(x)

loglike1 = sum(norm.logpdf(x, muhat1, sigmahat1))

print(loglike1)

-146.61679619684955
```

Now assume that a model is fit under the assumption that $\mu = 0$. Hence, only σ^2 is estimated.

```
In [73]: sigmahat2 = np.sqrt(np.mean(x**2))
loglike2 = sum(norm.logpdf(x, 0, sigmahat2))

print(loglike2)

-146.91962776341302
```

We can compare the values of AIC for the two fits.

```
In [74]: print(-2*loglike1 + 4)
print(-2*loglike2 + 2)

297.2335923936991
295.83925552682604
```

Also consider testing the null hypothesis $H_0 : \mu = 0$ versus the alternative $H_1 : \mu \neq 0$.

```
In [77]: semuhat1 = sigmahat1/10

WaldTestStat = muhat1/semuhat1
pvalue = 2*(1-norm.cdf(abs(WaldTestStat)))
print(pvalue)

0.43573028952293535
```

3. We observe that the log likelihood for the model which includes μ is larger than the log likelihood for the model which does not include μ as an unknown parameter. Which of the following is true?

Choose One:

- (a) It was certain to be the case that this log likelihood would be larger, so there is no conclusion can be drawn from this.
- (b) This is evidence that the model that includes μ is a better choice for this data set.
- (c) There must be a calculation error or a bug in the code.

4. We observe that for the model which assumes $\mu = 0$, the MLE for σ is calculated as

$$\hat{\sigma} = \sqrt{\frac{1}{n} \sum_i X_i^2}$$

This calculation takes advantage of what property of the MLE?

Choose One:

- (a) Approximate normality of the MLE.
- (b) Unbiasedness of the MLE.
- (c) The invariance of the MLE.
- (d) The delta method.

5. Suppose that $X_1, X_2, \dots, X_n$ are iid and each has pdf

$$f_X(x; \theta) = \frac{1}{2\theta} \exp\left(-\frac{|x|}{\theta}\right) \quad \text{for } -\infty < x < \infty \text{ and } \theta > 0.$$

Find the MLE for θ .

6. Suppose that $X_1, X_2, \dots, X_n$ are iid, each having the Geometric(p) distribution. See the table at the back of the exam for details on this distribution. Assume that $0 < p < 1$.

You can use each of the following without proof:

- The joint pmf for $\mathbf{X} = (X_1, X_2, \dots, X_n)$ for fixed $0 < p < 1$ is

$$\prod_{i=1}^n p(1-p)^{x_i} = p^n (1-p)^{\sum_i x_i}.$$

- The second derivative of $\log f_X(x; p)$ with respect to p is

$$\frac{\partial^2}{\partial p^2} (\log(p(1-p)^x)) = -\frac{x}{(1-p)^2} - \frac{1}{p^2}.$$

- The MLE for p is

$$\hat{p} = \left(\frac{n}{n + \sum_{i=1}^n X_i} \right).$$

- A useful derivative:

$$\frac{\partial}{\partial p} \left(\frac{1-p}{p} \right) = -\frac{1}{p^2}$$

Now, do each of the following:

- (a) Use an appropriate approximation to show that the standard error of the MLE $\hat{p}$ is approximately

$$\sqrt{\frac{p^2(1-p)}{n}}.$$

In practice, of course, we'd have to report the approximate standard error as

$$\sqrt{\frac{\hat{p}^2(1-\hat{p})}{n}}.$$

- (b) Using an appropriate approximation, construct a $100(1-\alpha)\%$ confidence interval for

$$E(X_i) = \left(\frac{1-p}{p} \right).$$

7. Let $X_1, \dots, X_n$ be a i.i.d. sample from a Bernoulli(p) distribution.

- Find the *Maximum Likelihood Estimator* (MLE) of p , $\hat{p}$, and its expected value.
- Compute the exact as well as the asymptotic variance of $\hat{p}$. How do they compare?
- A quantity that is sometimes of interest in binomial problems is the *log-odds ratio* (or simply, log-odds) given by: $\log \frac{p}{(1-p)}$.
 - What is the MLE of the log-odds. State clearly any property/fact that you're using.
 - Find the asymptotic variance of the MLE you found in (i).
 - Suppose that in the next 10 time periods, we have 5 increases in the stock price. Use this information and part and (i) and (ii) above to estimate the log-odds with an approximate 95% confidence interval.

Useful to know: $X_1, X_2, \dots, X_m$ i.i.d Bernoulli(p) $\implies \sum_{i=1}^m X_i \sim \text{Bin}(m, p)$.

8. Let $X_1, X_2, \dots, X_n$ be i.i.d $N(0, \sigma^2)$.

- (a) Find a method of moments estimator for σ^2 . Call it $\hat{\sigma}_1^2$.
- (b) Is $\hat{\sigma}_1^2$ unbiased?
- (c) A different estimator is suggested: $\hat{\sigma}_2^2 = \frac{1}{n} \left(\sum_{i=1}^n X_i \right)^2 = n \bar{X}^2$. Is $\hat{\sigma}_2^2$ unbiased?

9. Let $X_1, X_2, X_3, \dots, X_n$ be i.i.d. from the following distribution:

$$f_X(x; \theta) = \begin{cases} \theta e^x e^{-\theta e^x} & -\infty < x < \infty \\ 0 & \text{otherwise} \end{cases}$$

Show that the MLE for θ is

$$\hat{\theta}_{mle} = \frac{n}{\sum_{i=1}^n e^{X_i}}$$

10. Let $X_1, X_2, X_3, \dots, X_n$ be i.i.d. from the following distribution:

$$f_X(x; \theta) = \begin{cases} \frac{2x}{\theta} e^{-\frac{1}{\theta} x^2} & x > 0 \\ \theta > 0 \\ 0 & \text{otherwise} \end{cases}$$

Useful to know:

- If $X \sim f_X(x; \theta)$, then $X^2 \sim \exp(1/\theta)$.

- (a) What is the MLE for θ ?
- (b) What is the MLE for $P(X > t)$? Justify your answer.
- (c) What is the asymptotic variance of $\hat{\theta}_{MLE}$?
- (d) Derive an approximate 95% confidence interval for θ based on the MLE.
- (e) $g(\theta) = \sqrt{\theta}$ is of particular interest since if $X \sim f_X(x; \theta)$, then $E(X) = \frac{\sqrt{\pi\theta}}{2}$. What is the MLE for $\sqrt{\theta}$? Is it unbiased?
- (f) What is the asymptotic variance of the MLE you found in the previous part?

11. Suppose that $X_1, X_2, \dots, X_n$ are i.i.d. with each having probability density function (pdf)

$$f(x; \theta) = \theta e^x e^{-\theta e^x}, \quad -\infty < x < \infty, \quad \theta > 0.$$

- (a) Find the maximum likelihood estimator (MLE) for θ .
(You should verify that you are actually finding a maximum.)

Guidance: Note that

$$\prod_{i=1}^n e^{-\theta e^{x_i}} = e^{-\theta \sum_{i=1}^n e^{x_i}}$$

- (b) What is the approximate standard error of the MLE for θ ?

12. Suppose that $X_1, X_2, \dots, X_n$ are i.i.d. from the Poisson(λ) distribution. A couple facts that we showed in lecture:

- The maximum likelihood estimator (MLE) for λ is the sample mean $\bar{X}$.
- The Fisher information in this case is $I(\lambda) = 1/\lambda$.

- (a) Is the MLE an unbiased estimator of λ in this case?

(CHOOSE ONE)

- i. Yes.
- ii. No.
- iii. Whether or not it's unbiased depends on the value of n .
- iv. Whether or not it's unbiased depends on the value of λ .

- (b) What is the mean squared error (MSE) of this estimator of λ ?

(Your answer can be a function of λ and n .)

- (c) Suppose a sample of size $n = 50$ is observed, with $\bar{x} = 7.2$. Construct an approximate 95% confidence interval for λ .

13. Suppose that a sample of size $n = 100$ is taken, and you are willing to model it as i.i.d. draws from a normal distribution with mean μ and variance σ^2 . The sample variance is $s^2 = 79.3$. Construct a 90% confidence interval for σ^2 .
14. Suppose that $X_1, X_2, \dots, X_n$ are i.i.d. $\text{Normal}(\mu, \sigma^2)$, with both μ and σ^2 unknown. What is the MLE of μ/σ ?
15. Suppose that $X_1, X_2, \dots, X_n$ are i.i.d. $\text{Normal}(\mu, \sigma^2)$, with both μ and σ^2 unknown. We saw in class that the MLE for σ^2 has variance $2\sigma^4/n$. Approximate the variance of the MLE for σ .